

Interface Intelligence Operating Blueprint

Recommended product definition, transaction-lineage model, agent operating system, API surface, and branded output architecture for revenue, customer, and member lifecycle evidence.

681 QTR SCENARIO RECORDS REVIEWED	47 CONTROL PATTERNS IN SOURCE WORKBOOK	3 PRIMARY LIFECYCLE TOKENS	13 CORE AGENT ROLES	1 INTERFACE EVIDENCE LEDGER
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Recommended Name	HandoverOS Interface Intelligence Layer
Definition	A governed transaction-lineage layer that sits above client integrations, APIs, imports, exports, and system-to-system connectors. It turns every interface event into an agent-readable evidence record with values, transformations, waterfall steps, revenue/customer/member tokens, controls, confidence, contribution attribution, and recommended action.

Prepared from Gmail connector analysis, attached Salt Basin design references, existing repo lineage code, and relevant Salt Basin Website thread context.

1. Product Doctrine

Most revenue leakage and customer-friction issues are not created where they become visible. They are created upstream in a field, rule, approval, transform, stale source, missing handoff, missing token, or control gap that no one can see later.

Interface Intelligence makes those hidden moments visible. It does not only ask whether data moved. It asks what changed, why it changed, which lifecycle object it affected, which downstream processes now depend on it, which control should have caught it, and whether the business can trust the result.

Product One-Liner

HandoverOS Interface Intelligence traces every revenue, customer, and member lifecycle data transaction across systems, then uses Salt Basin agents to explain lineage, detect mismatches, identify hidden operational risk, reconcile contribution, and preserve reusable institutional memory.

Strategic Fit

- Evidence layer under HandoverOS Q2R diagnostics.
- Governed context layer for BestyStaff and future Salt Basin agents.
- High-value use case for the Salt Basin Contribution Intelligence API.
- Natural extension of the current repo's field-level lineage and snapshot capability.

Public-Safe Boundary

Public outputs can show maps, scorecards, governance checklists, lineage diagrams, capability taxonomies, and example schemas. They should not expose raw Gmail-derived materials, proprietary workbook internals, client/employer artifacts, private formulas, or full field taxonomies without approval.

2. Three Primary Token Model

The product needs three linked primary tokens because revenue truth, customer truth, and member/entitlement truth travel together but are not the same thing.

Token	Primary Question	Lifecycle
Revenue Lifecycle Token	What commercial or financial object is this event changing, and where will it land downstream?	Prospect -> Pipeline -> Proposal -> Contract -> Order -> Subscription -> Bill -> Invoice -> Collect -> Recognize -> Renew/Expand -> Adjust
Customer Journey Token	What customer-adjacent moment is this event changing, and what experience or retention consequence follows?	External Lead -> Qualified Prospect -> Negotiation -> Signature -> Onboarding -> Active Customer -> Payment -> Support/Change -> Renewal/Expansion/Churn
Member Journey Token	Which organization, site, member, end user, covered population, or free-tier user is entitled, activated, supported, billed, renewed, or changed?	Eligible Population -> Invited Member -> Entitled Member -> Activated User/Site -> Licensed Use -> Support/Utilization -> Expansion/Change -> Renewal/Termination

Design Rule

The tokens are linked but not collapsed. A payer can sign the contract while onboarding, fulfillment, utilization, support, and renewal happen at a child organization, related organization, site, provider group, employer population, plan member group, individual member, or end-user level.

Canonical Entities

Domain	Entities
GTM	Lead, Account, Contact, Opportunity, Quote, Proposal
Commercial	Contract, Performance Obligation, Order, Subscription, Product, Price, Discount, Approval
Finance	Invoice, Payment, Collection, Revenue Schedule, ARR, NRR, Adjustment
Customer	Onboarding, Entitlement, Usage, Case, Service Request, Change Request, Renewal, Churn
Member	Parent Org, Child Org, Site, Provider, Employer Group, Plan Member, Individual Member, End User, Free Tier

3. Token Chain Dimensional Model

The expanded model links commercial promise, member entitlement, financial execution, and resource contribution into one chain that can be traced, scored, and remembered.

Layer	What It Captures
Deal Contribution Intelligence	deal type, valuation size, investment type, industry vertical, actively seeking criteria match, board skills, sponsor priorities, value-creation thesis
Contract Economics	payer, bill-to, invoice destination, payment method, remittance method, annual recurrence, usage pricing, discounts, variable consideration, support terms, renewal notice
Member Coverage	parent organization, child organization, site, provider, employer population, plan member population, individual license, free tier, activation, utilization, support, churn/renewal status
Stage Gate IO	required inputs, outputs, approvals, blockers, decision owner, output artifacts, resource contributors, and technology contributors
Resource Contribution	human role, user type, team, system, AI agent, workflow, technology, cost basis, elapsed time, quality score, and ROI attribution
Confidence	input coverage, output coverage, transformation confidence, token match confidence, financial reconciliation confidence, and human/agent review status

Chain

Deal Type -> Customer/Payer -> Contract Obligation -> Revenue Lifecycle Token -> Customer Journey Token -> Member Journey Token -> Stage Gate -> Interface Transaction -> Field Lineage -> Financial Transaction -> Contribution Event -> Confidence Reconciliation -> Memory Candidate.

4. Interface Evidence Ledger

Every interface transaction should become a durable evidence record that a human or agent can query later. The ledger should support both real-time API capture and batch ingestion from exports, imports, ETL jobs, and historical files.

Area	Captured Data
Identity	transaction id, correlation id, idempotency key, connector id, environment, tenant, timestamp
Direction	source system, target system, source object, target object, operation, trigger
Payload	raw payload reference, masked payload summary, before/after values, schema version
Transformation	rule id, mapping version, formula, enrichment source, fallback/default logic
Waterfall	ordered steps, duration, status, retry count, actor, system response
Lifecycle	revenue lifecycle token, customer journey token, member journey token, supporting agent tokens, stages, canonical entity ids
Contract/Entitlement	payer, bill-to, fulfill-to, covered organization/population, performance obligations, pricing, support, renewal, usage, payment, remittance, invoice destination
Governance	evidence status, control policy, sensitivity level, retention policy, approval state
Agent Output	summary, anomaly flags, root-cause hypothesis, contribution breakdown, confidence reconciliation, recommended action

Implementation Note The repo already has field_lineage and data_snapshots. Interface Intelligence should extend that foundation with transaction identity, connector identity, lifecycle tokens, control evidence, and agent analysis.

5. Data Model

Table	Purpose
interface_connectors	Registered systems, provider, tenant, auth scope, environment, owner, and status.
interface_transactions	One durable API call, webhook, import, export, sync job, or manual upload event.
interface_transaction_steps	Ordered waterfall steps: extract, map, transform, validate, enrich, write, acknowledge, retry.
interface_field_lineage	Field movement, prior value, transformed value, rule id, confidence, hash, and masking policy.
lineage_tokens	Revenue Lifecycle, Customer Journey, Member Journey, and supporting agent-token records.
token_edges	Relationships among lead, account, payer, employer, plan, site, member, user, quote, contract, order, subscription, invoice, payment, case, and change request.
member_identity_links	Links payer, customer, parent org, child org, site, provider, employer group, plan, member, end user, and free-tier identity records.
member_entitlements	Covered population, license type, eligibility, activation, utilization, support coverage, access tier, and renewal/change status.
token_stage_gates	Required inputs, outputs, decisions, approvals, and evidence for each token stage gate.
contribution_events	Human, user-type, team, system, AI agent, workflow, and technology contribution records.
confidence_reconciliations	Conflicting evidence, coverage scores, confidence scores, review status, and final resolution.
interface_controls	Required controls and policies by connector, object, lifecycle stage, or field class.
control_executions	Evidence that a control ran, passed, failed, was skipped, or required human review.
agent_transaction_analyses	Agent explanations, risks, root-cause hypotheses, confidence, and recommendations.
memory_candidates	Reusable insight candidates awaiting classification and approval.
usage_meter_events	Salt Basin metering event at the moment of billable capability consumption.

6. API Surface

The API should align with the existing Contribution Intelligence API architecture: versioned endpoints, metered billable capability calls, connector validation, client-managed model options, and durable usage events.

Endpoint	Purpose	Billable
POST /api/v1/interface-intelligence/events	Capture one transaction event.	Conditional
POST /api/v1/interface-intelligence/events/batch	Capture batch imports, ETL jobs, or sync windows.	Conditional
GET /api/v1/interface-intelligence/transactions/:id	Retrieve transaction forensic record.	No
GET /api/v1/interface-intelligence/tokens/:tokenId/graph	Retrieve revenue, customer, member, and supporting-token graph.	No or premium by depth
POST /api/v1/interface-intelligence/analyze/transaction	Run agent analysis on one transaction.	Yes
POST /api/v1/interface-intelligence/analyze/token	Run cross-system analysis for a token.	Yes
POST /api/v1/interface-intelligence/connectors/validate	Validate connector access, objects, scopes, and rate limits.	Conditional
POST /api/v1/pricing/estimate	Estimate usage, volume, and advisory scope.	No

Security Rule

Do not inherit client system security as Salt Basin's security model. Salt Basin needs tenant auth, scoped API keys or OAuth client credentials, RBAC, object-level authorization, audit logs, webhook signatures, rate limits, idempotency keys, encryption, retention controls, and sandbox/production separation.

7. Agent Operating Model

Agent loop: Observe -> Classify -> Trace -> Explain -> Detect -> Recommend -> Remember.

Agent	Job	Max Volume Per Worker
Interface Transaction Analyst	Inspect API calls, imports, exports, and sync events for missing values, errors, drift, retries, and mismatches.	250 transactions or 50 error clusters
Integration Mapping Agent	Map source/target objects, fields, transformations, dependencies, and failure points.	150 fields, 20 objects, or 10 payloads
Revenue Lifecycle Tracer	Follow the Revenue Lifecycle Token across the commercial and financial lifecycle.	100 tokens or 500 related events
Customer Journey Tracer	Follow the customer-adjacent token from lead through service, change, renewal, churn.	100 tokens or 500 related events
Member Journey Tracer	Follow org members, child orgs, sites, individual members, end users, free-tier users, and entitlement coverage.	100 tokens or 500 related events
Contract Obligation Mapper	Extract performance obligations, payer responsibility, billing triggers, renewal terms, support, discounts, usage, payment, and variable consideration.	75 contracts or 250 clauses
Financial Transaction Reconciler	Trace payment, remittance, invoice, adjustment, collection, and revenue recognition events to the Revenue Lifecycle Token.	250 financial events
Contribution Intelligence Agent	Attribute human, user-type, team, system, AI agent, workflow, and technology contributions to stage gates and outcomes.	200 contribution events
Confidence Reconciliation Agent	Compare conflicting token evidence, score coverage/confidence, and route human review.	150 conflicts or 500 scored fields
Control Evidence Agent	Check whether required controls ran, passed, failed, or require human review.	50 controls or 250 executions
Root Cause Pattern Agent	Cluster recurring failures into leakage patterns, governance causes, and process defects.	300 anomalies or 25 clusters
Benchmark And Claims Guard	Prevent stale, unsupported, or external claims from entering public/client outputs.	100 claims
Memory Steward	Classify reusable memory candidates as public-safe, internal, restricted, confidential, or quarantine.	50 sensitive or 250 low-risk

8. Parallel Agent Sizing

Sizing Formula worker_count = ceiling(total_units / max_units_per_worker). functional_lead_count = ceiling(worker_count / 5). management_agent_count = ceiling(functional_lead_count / 3).		
Size	Practical Scope	Suggested Agent Shape
Small	Up to 100 pages, 1k rows, 50 defects, or one process tower.	1-2 workers, 0-1 lead, 1 manager/orchestrator
Medium	100-500 pages, 1k-25k rows, 50-300 defects, or 3-5 towers.	3-6 workers, 1-2 leads, 1-2 managers
Large	500-2000 pages, 25k-250k rows, 300-1500 defects, or 5-10 towers.	8-20 workers, 3-5 leads, 2-4 managers
Portfolio	Multiple companies or 2000+ pages, 250k+ rows, 1500+ defects.	20-60 workers, 5-12 leads, 4-8 managers

Recommended default for first production client: one manager/orchestrator, one integration lead, one revenue lifecycle lead, one member journey tracer, two transaction analysts, one contract obligation mapper, one contribution intelligence agent, one control evidence agent, one memory steward, and one claims guard on call for any external-facing output.

9. Salt Basin Branded Outputs

Output	Audience	Purpose
Executive Interface Intelligence Brief	Sponsor, CFO, CRO, COO, CTO, PortOps	Explain the hidden operating truth and recommended action.
Transaction Forensic Packet	RevOps, IT, Finance Ops, Data	Show what happened in a transaction, field by field and step by step.
Token Lineage Graph	Product, RevOps, Finance, CS	Trace revenue and customer truth across systems.
Member Entitlement Graph	Product, CS, Finance, Healthcare/Market Ops	Trace payer, parent org, child org, site, member, user, free-tier, and entitlement relationships.
Contract Obligation Matrix	Legal, Finance, RevOps, Product	Show obligations, billing triggers, payment responsibility, support terms, renewal terms, discounts, and variable consideration.
Contribution Chain Scorecard	Sponsors, PortOps, Transformation Office	Compare human and technology contribution cost basis, ROI, stage order, contributor combinations, and deal intelligence.
Control Evidence Matrix	Finance, Audit, Security, Transformation Office	Show controls run, gaps, skips, and review items.
Root Cause Heatmap	Transformation leaders	Prioritize recurring failure patterns by impact, frequency, and confidence.
Source Delta Card	Research and product team	Convert new evidence into model updates without overclaiming.
Memory Candidate Register	Internal Salt Basin agents	Preserve reusable insight under classification rules.

Design System Fit

Use navy for authority, gold for executive emphasis, teal for data and AI, green for outcomes, plum for risk or strategic emphasis. Favor KPI tiles, outcome cards, metadata chips, lineage rivers, evidence matrices, and validation checklists.

10. Build Path

Phase	Build
Now	Define schema, add transaction tables, capture API events, render forensic view, attach revenue/customer/member tokens.
Next	Run agent analyses for transaction anomalies, token trace, member entitlement trace, contract obligation trace, contribution breakdown, and control evidence.
Then	Add connector profiling, pricing estimate, usage metering, public-safe output templates, and client portal views.
Later	Add continuous research refresh, benchmark diffing, source delta cards, and portfolio-level intelligence views.

Current Confidence

Area	Confidence	Reason
Internal HandoverOS/QTR shape	High	Gmail connector research, workbook attachment reads, repo memory docs, and current lineage code converge.
Technical feasibility in repo	High	Existing field lineage and snapshots provide a base to extend.
External benchmark claims	Medium until refreshed	Active memory requires current authoritative refresh before public/client use.
Exact public naming	Medium	Recommended name is strong but can be refined with Betsy's positioning preference.